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$$11). x > 0 \quad \cancel{Wz} + W = z - 1 \Rightarrow z = \frac{-W-1}{W-1} = \frac{1+W}{1-W}$$

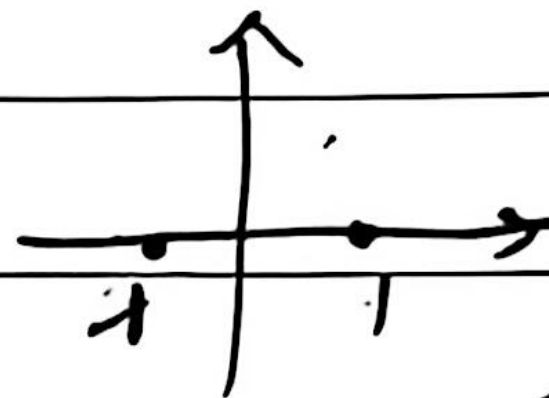
$$= \frac{1+u+iv}{1-u-iv} = \frac{(1+u+iv)(1-u+iv)}{(1-u)^2+v^2} = \frac{1-u^2-v^2+2vi}{(1-u)^2+v^2} = x+iy$$

$$\begin{cases} x = \frac{1-u^2-v^2}{(1-u)^2+v^2} > 0 \Rightarrow u^2+v^2 < 1 \\ y = \frac{2v}{(1-u)^2+v^2} \end{cases} \quad \{w; |w| < 1\}$$

$$|z|^2 = \frac{(1-u^2-v^2)^2 + 4v^2}{((1-u)^2+v^2)^2} < 1 \Rightarrow (1-u^2)^2 - 2(1-u^2)v^2 + v^4 + 4v^2 < (1-u^2)^2 + 2v^2(1-u)$$

$$(2). |z| = \frac{|w+1|}{|w-1|} < 1 \quad |w+1| < |w-1|$$

$$\therefore \operatorname{Re} w < 0$$



$$16.12) W_1 = z^4 \Rightarrow D_1 = \{W_1; 0 < \arg w_1 < \pi\}$$

$$W_2 = \frac{W_1 - i}{W_1 + i} \Rightarrow D_2 = \{W_2; |W_2| < 1\}$$

$$\therefore W = \frac{z^4 - i}{z^4 + i}$$

$$(3) ① W_1 = \left(\frac{z}{2}\right)^4 \Rightarrow D_1 = \{W_1; 0 < \arg w_1 < \pi, |W_1| < \frac{1}{16}\}$$

$$② W_2 = \frac{W_1 + 1}{1 - W_1} \Rightarrow D_2 = \{W_2; 0 < \arg w_2 < \frac{\pi}{2}\}$$

$$③ W_3 = W_2^2 \Rightarrow D_3 = \{W_3; \operatorname{Im} W_3 > 0\}$$

$$④ W = \frac{W_3 - i}{W_3 + i} \rightarrow D_4 = \{W; |W| < 1\}$$

$$\therefore W = \frac{\left(\frac{(\frac{z}{2})^4 + 1}{1 - (\frac{z}{2})^4}\right)^2 - i}{\left(\frac{(\frac{z}{2})^4 + 1}{1 - (\frac{z}{2})^4}\right)^2 + i}$$



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$$\textcircled{1} w_1 = z^2 \quad D_1: \{w_1; 0 < w_1 < \pi\}$$



$$1+i \rightarrow 2i$$

$$0 \rightarrow 0$$

$$\textcircled{2} w = k \frac{w_1 - 1 - i}{w_1 - 1 + i} \quad D_2: \{w_1; |w_1| < 1\}$$

$$\therefore \left(w = k \frac{z^2 - 1 - i}{z^2 - 1 + i} \right) \quad \begin{aligned} w(1+i) &= k \frac{1}{3} = 0 \\ w(0) &= -k = 1 \end{aligned}$$

$$w = k \frac{z^2 - 1 - i}{z^2 - 1 + i} \quad w(0) = -k = 1 \quad k = -1$$

$$\Rightarrow w = \frac{z^2 - 1 - i}{z^2 - 1 + i}$$

$$18. \quad w = k \frac{z - \alpha}{z - \beta}$$

$$w(-i) = 0$$

$$w(i) = \infty$$

$$w = e^{i\theta} \frac{z + i}{z - i}$$

$$w(1) = e^{i\theta} \frac{1+i}{1-i} = e^{i(\theta + \frac{\pi}{2})}$$

$$\arg w(1) = \varphi_0 \Rightarrow \theta = \varphi_0 - \frac{\pi}{2}$$

$$\therefore w = e^{i(\varphi_0 - \frac{\pi}{2})} \frac{z + i}{z - i}$$

$$-\frac{3}{2}\pi$$

$$19. \quad \textcircled{1} w_1(0) = 0 \quad w_1(z) = \infty$$

$$w_1 = \frac{w}{w-2}$$

$$w_1(1+i) = \frac{1+i}{i-1} = \frac{2i}{-2} = -i$$

$$w_1(-2) = \frac{-2}{-4} = \frac{1}{2}$$

$$w_1(i) = \frac{i}{i-2} = \frac{i(i+2)}{-5} = \frac{-1+2i}{-5} = \frac{1}{5} - \frac{2}{5}i$$

$$\therefore D_1: \{w_1, \operatorname{Re} w_1 > 0, \operatorname{Im} w_1 < 0\}$$

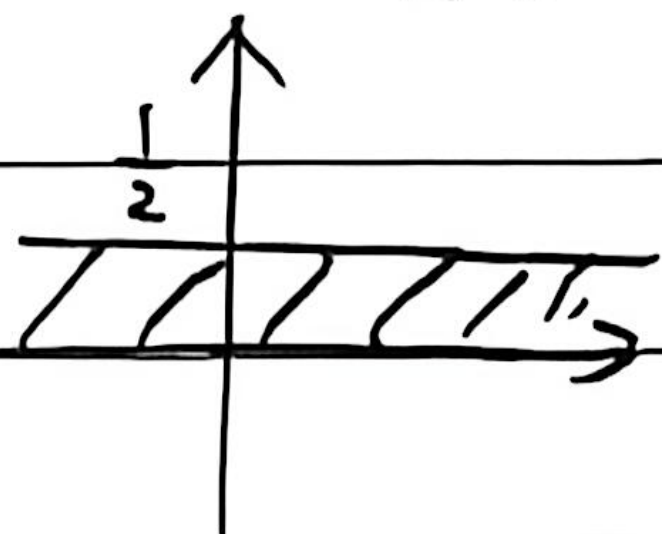
$$\textcircled{2} w_2 = \frac{w-1}{w+1}$$



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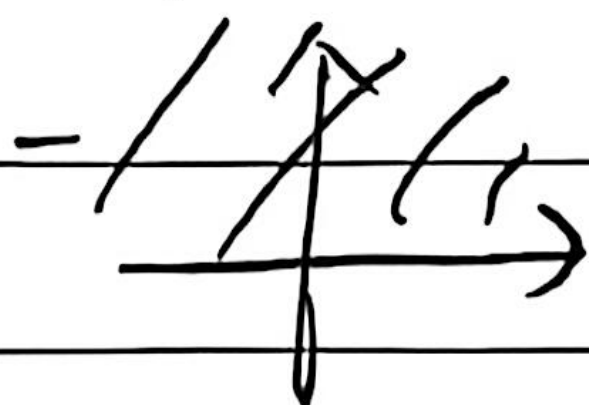
(1) ① $w_1 = \frac{z-2}{z+2}$

$w_1(z) = \infty$ $w_1(0) = 0$ $w_1(-2) = \frac{1}{2}$



$D_1: \{w_1, 0 < \text{Im} w_1 < \frac{1}{2}\}$

② $w = e^{2\pi w_1} = e^{2\pi \frac{z}{z+2}}$

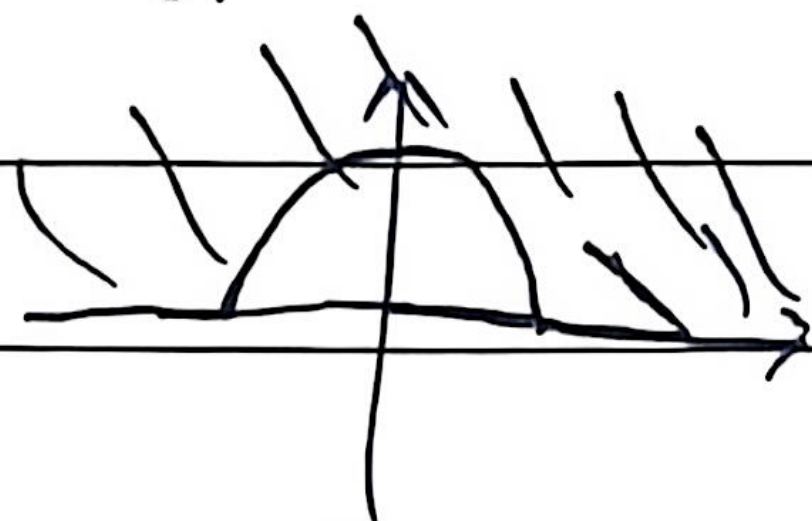


① $w_1 = \begin{cases} z, & \text{Re } z > 0 \\ -\bar{z}, & \text{Re } z < 0 \end{cases}$

$D_1: \{w_1, 0 < \text{Im} w_1 < \alpha\}$

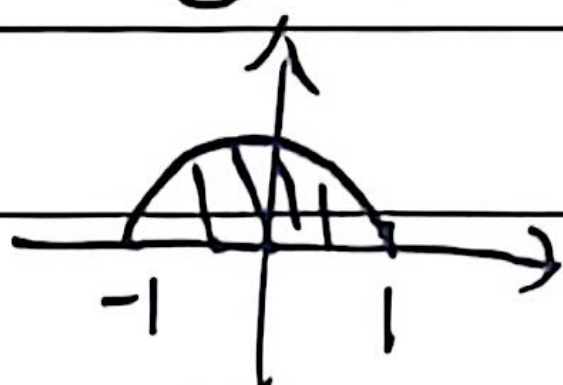
② $w = e^{\frac{2}{\alpha} w_1} = \begin{cases} e^{\frac{2}{\alpha} z}, & \text{Re } z > 0 \\ e^{\frac{2}{\alpha} \bar{z}}, & \text{Re } z < 0 \end{cases}$

or: ① $w_1 = e^{\frac{2}{\alpha} z}$



④ $w_4 = w_3^2$

② $w_2 = \frac{1}{w_1}$



③ $w_3 = \frac{w_2+1}{w_2-1}$



$w_3(0) = 1$ $w_2(i) = \frac{i+1}{i-1} = \frac{2i}{-2} = -i$

